# DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW

# EVALUATION SCHEME & SYLLABUS FOR

## B. TECH. III YEAR

BIOMEDICAL ENGINEERING

## ON

## CHOICE BASED CREDIT SYSTEM (CBCS)

## [Effective from the Session: 2018-19]

# Syllabus & Evaluation Scheme

## Year III, Semester V

| Sr. No. | Sub Code | Subject Name | L-T-P | Th/Lab Marks | Sessional | | Total | Credit |
|---|---|---|---|---|---|---|---|---|
| | | | | ESE | CT | TA | | |
| 1 | RAS501 | Managerial Economics | 3--0--0 | 70 | 20 | 10 | 100 | 3 |
| 2 | RAS502 /RUC501 | Sociology/Cyber Security | 3--0--0 | 70 | 20 | 10 | 100 | 3 |
| 3 | RBM501 | Control System-I | 3--0--0 | 70 | 20 | 10 | 100 | 3 |
| 4 | RBM502 | Bioinstrumentation | 3--1--0 | 70 | 20 | 10 | 100 | 4 |
| 5 | RBM503 | Microprocessor & Application | 3--0--0 | 70 | 20 | 10 | 100 | 3 |
| 6 | REC501, REC052, RBM051, REC054 | Deptt. Elective Course 1 | 3--1--0 | 70 | 20 | 10 | 100 | 4 |
| 7 | RBM551 | Control System Lab | 0--0--2 | 50 | | 50 | 100 | 1 |
| 8 | RBM552 | Bioinstrumentation Lab - I | 0--0--2 | 50 | | 50 | 100 | 1 |
| 9 | RBM553 | Microprocessor Lab -I | 0--0--2 | 50 | | 50 | 100 | 1 |
| 10 | RBM554 | Integrated Circuit Lab | 0--0--2 | 50 | | 50 | 100 | 1 |
| | TOTAL | | | 620 | 120 | 260 | 1000 | 24 |

**DEPTT. ELECTIVE COURSE-1**

1. **REC501 - Integrated Circuit**
2. **REC052 - Computer Architecture and Organization**
3. **RBM051 - Human Anatomy &Physiology**
4. **REC054 - Artificial Neural Network**

# Syllabus & Evaluation Scheme

## Year III, Semester VI

| Sr. No. | Sub Code | Subject Name | L-T-P | Th/ LAB Mark | Sessional | | Total | Credit |
|---|---|---|---|---|---|---|---|---|
| | | | | ESE | CT | TA | | |
| 1 | RAS601 | Industrial Management | 3--0—0 | 70 | 20 | 10 | 100 | 3 |
| 2 | RAS602 / RUC601 | Sociology /Cyber Security | 3--0—0 | 70 | 20 | 10 | 100 | 3 |
| 3 | RBM-601 | Microcontroller & Biomedical Applications | 3--0—0 | 70 | 20 | 10 | 100 | 3 |
| 4 | RBM-602 | Biomedical Signal Processing | 3--1—0 | 70 | 20 | 10 | 100 | 4 |
| 5 | RBM-603 | Biomaterials | 3--0—0 | 70 | 20 | 10 | 100 | 3 |
| 6 | RBM-061 RBM-062 RBM-063 RBM-064 | Deptt. Elective Course-2 | 3--1—0 | 70 | 20 | 10 | 100 | 4 |
| 7 | RBM-652 | Biomedical Signal Processing Lab | 0--0—2 | 50 | | 50 | 100 | 1 |
| 8 | RBM-651 | Bioinstrumentation Lab - II | 0--0—2 | 50 | | 50 | 100 | 1 |
| 9 | RBM-653 | Microprocessor Lab - II | 0--0—2 | 50 | | 50 | 100 | 1 |
| 10 | RBM-654 | Seminar | 0--0—2 | 50 | | 50 | 100 | 1 |
| | TOTAL | | | 620 | 120 | 260 | 1000 | 24 |

**DEPTT. ELECTIVE COURSE-2**

1. **RBM-061 Laser & Fibre Optics & Its Medical Applications.**
2. **RBM-062 Bioelectricity**
3. **RBM-063 Biomechanics**
4. **RBM-064 Physiological Control System & Simulation Modeling**

| RBM501 | CONTROL SYSTEM- I | L T P: 300 |
|---|---|---|

Unit-I

Basic Components of a control system, Feedback and its effect, types of feedback control systems.Block diagrams and signal flow graphs, Modeling of Physical systems: electrical networks, mechanical systems elements, equations of mechanical systems, sensors and encoders in control systems, DC motors in control systems.

Unit-II

State-Variable Analysis: Vector matrix representation of state equation, state transition matrix, state-transition equation, relationship between state equations and high-order differential equations, relationship between state equations and transfer functions.

Unit-III

**Time domain Analysis of Control Systems:** Time response of continuous data systems, typical test signals for the time response of control systems, the unit step response and time-domain specifications, Steady-State error, time response of a first order system, transient response of a prototype second order system.

Unit-IV

**Stability of Linear Control Systems**: Bounded-input bounded-output stability-continuous data systems, zero-input and asymptotic stability of continuous data systems, methods of determining stability, Routh Hurwitz criterion.

Unit-V

**Frequency Domain Analysis:** $M_r$ (resonant peak) and $\omega_r$ (resonant frequency) and bandwidth of the prototype Second order system, effects of adding a zero to the forward path, effects of adding a pole to the forward path, Nyquist stability criterion, relative stability: gain margin and phase margin, stability analysis with the Bode plot.

**Text Book:**

B.C. Kuo & Farid Golnaraghi, “Automatic Control Systems”, 8th Edition, Wiley.

**Reference Books:**

William A. Wolovich, “Automatic Control Systems”, Oxford University Press,2010. Series, 3rd Edition, Tata McGraw Hill, Special Indian Edition 2010.

| RBM502 | BIOINSTRUMENTATION | L T P: 310 |
|---|---|---|

UNIT I

Transducers - Classification, selection of transducers, circuit based on transduction. Temperature transducers – Displacement transducer - Pressure transducer - catheter tip transducers. Photoelectric transducers - Flow transducers - Piezoelectric transducers and their applications, Biological receptors and receptor characteristics.

UNIT II

Reference electrodes, The $pO_2$ electrodes, Membrane electrodes, Blood gas analysis, Transcutaneous $pO_2$ and $pCO_2$ transducers, Fiber optic chemical transducer, Ion specific electrodes, Ionic content of blood, ISFET for glucose, urea.

UNIT III

Electrocardiogram (ECG), Electroencephalogram (EEG), Electromyogram (EMG), Electrooculogram (EOG), Electroretinogram (ERG), Recording Electrodes – Electrode-tissue interface, polarization, skin contact impedance, motion artifacts, Silver-Silver Chloride electrodes, Electrodes for ECG, Electrodes for EEG, Electrodes of EMG, Electrical conductivity of electrode jellies and creams, microelectrodes, Needle electrodes.

UNIT IV

Principle of dialysis, Artificial kidney, Types, function and working of dialysis, Performance analysis of dialysis, Membranes used for hemodialysis, Block diagram and working of hemodialysis machine, Portable kidney machine – working and flow diagram.

UNIT V

Physiological effects of electrical currents, macroshock and microshock, preventive measures to reduce shock hazards, Leakage current, isolation of patient circuits, safety of electrically susceptible patients, radiation hazards and safety, shielding, open ground problem and earthing methods.

**TEXT BOOKS:**


1. R. S. Khandpur, Biomedical Instrumentation Technology and Applications, McGraw-Hill Professional, 2004
2. Leslie Cromwell, Fred. J. Weibell and Erich. A. Pfeiffer, “Biomedical Instrumentation and Measurements”, 2nd Edition, PHI, 2003. (UNIT I, III)
3. John G. Webster, Medical Instrumentation: Application and Design, 3rd edition, John Wiley & Sons, New York, 1998. (UNIT II)
4. Raja Rao, C, Guha, S.K, Principles of Medical Electronics and Biomedical Instrumentation, Orient Longman Publishers (2000) (UNIT V)


**REFERENCES :**


1. R.Anandanatarajan, “Biomedical Instrumentation”, PHI Learning, 2009.
2. M. Arumugam, “Biomedical Instrumentation”, Anuradha Agencies Publishers, Vidayal Karuppar, 612 606, Kumbakonam, R.M.S: 1992

| RBM503 | MICROPROCESSOR & APPLICATION | L T P: 300 |
|---|---|---|

**UNIT1**

**THE 8085 PROCESSOR:**

Introduction to microprocessor, 8085 microprocessor: Architecture, instruction set.

**UNIT2**

**THE 8086 MICROPROCESSOR** :

Architecture, block diagram of 8086, memory segmentation and physical address computations, program relocation, addressing modes, Pin Diagram and description of various signals instruction formats, Instruction Set , Assembler instruction format, Directives and Operators

**UNIT3**

**INTERFACING DEVICE**:

The 8255 PPI chip: DMA CONTROLLER (8237),

**UNIT4**

**INTERRUPT AND TIMER:**

8259 Programmable interrupt controller, Programmable interval timer chips (8253/8254).

**TEXT BOOKS:**

1.Microprocessor Architecture, Programming & Applications with 8085: Ramesh S Gaonkar; Wiley Eastern Ltd.
2 The Intel Microprocessors 8086- Pentium processor: Brey; PHI

**REFERENCE BOOKS:**

1. Microprocessors and interfacing: Hall; TMH
2. The 8088 & 8086 Microprocessors-Programming, interfacing, Hardware & Applications : Triebel & Singh; PHI
3. Microcomputer systems: the 8086/8088 Family: architecture, Programming & Design: Yu-Chang Liu & Glenn A Gibson; PHI.
4. Advanced Microprocessors and Interfacing: Badri Ram; TMH

| RBM551 | CONTROL SYSTEM LAB | L T P: 002 |
|---|---|---|

1. DC SPEED CONTROL SYSTEM
   (a) To study D.C. speed control system on open loop and close loop.
   (b) To study of Transient performance, another time signal is added at the input of control Circuit.
   (c) To study how eddy current breaking is being disturbance rejected by close and open loop.
2. DC MOTOR POSITION CONTROL
   (a) To study of potentiometer displacement constant on D.C. motor position control.
   (b) To study of D. C. position control through continuous command.
   (c) To study of D.C. position control through step command.
   (d) To study of D.C. position control through Dynamic response.
3. AC MOTOR POSITION CONTROL
   (a) To study of A.C. motor position control through continuous command.
   (b) To study of error detector on A.C. motor position control through step command.
   (c) To study of A.C. position control through dynamic response.
4. MAGNETIC AMPLIFIER
   (a) To study Input / Output characteristic of a magnetic amplifier in mode (i) Saturable Reactor, (ii) Self Saturable Reactor.
5. SYNCHRO TRANSMITTER / RECEIVER
   (a) To study of Synchro Transmitter in term of Position v/s Phase and voltage magnitude with respect to Rotor Voltage Magnitude/Phase.
   (b) To study of remote position indication system using Synchro-transmitter/receiver.
6. PID CONTROLLER
   (a) To observe open loop performance of building block and calibration of PID Controls.
   (b) To study P, PI and PID controller with type 0 system with delay.
   (c) To study P, PI and PID controller with type 1 system.
7. LEAD LAG COMPENSATOR
   (a) To study the open loop response on compensator.
   (c) Close loop transient response.
8. LINEAR SYSTEM SIMULATOR
   (a) Open loop response
   (i) Error detector with gain, (ii) Time constant, (iii) Integrator
   (b) Close loop system
   (I) First order system (II) Second order system (III) Third order system
9. Introduction to MATLAB (Control System Toolbox), Implement at least any two experiment in MATLAB.
   a. Different Toolboxes in MATLAB, Introduction to Control Systems Toolbox.
   b. Determine transpose, inverse values of given matrix.
   c. Plot the pole-zero configuration in s-plane for the given transfer function.
   d. Determine the transfer function for given closed loop system in block diagram representation.
   e. Plot unit step response of given transfer function and find peak overshoot, peak time.
   f. Plot unit step response and to find rise time and delay time.
   g. Plot locus of given transfer function, locate closed loop poles for different values of k.
   h. Plot root locus of given transfer function and to find out S, Wd, Wn at given root & to discuss stability.
   i. Plot bode plot of given transfer function.
   j. Plot bode plot of given transfer function and find gain and phase margins
   k. Plot Nyquist plot for given transfer function and to compare their relative stability
   l. Plot the Nyquist plot for given transfer function and to discuss closed loop stability, gain and phase margin.

| RBM552 | BIOINSTRUMENTATION LAB-I | L T P: 002 |
|---|---|---|

**LIST OF EXPERIMENTS**

1. Study of pulmonary function analyzer using spirogram.
2. To study finger tip oximeter.
3. Designing of instrumentation amplifier.
4. Designing of notch filter.
5. To study voltage regulator IC 7805, 7809, 7812 series.
6. To determine Bradycardia and Tachycardia using ECG Training Kit.
7. To determine heart rate using ECG simulator Kit.
8. Circuitry explanation for patient leakage current.
9. To determine balancing condition for thermistor using wheat stone bridge.
10. Study of pressure changes using strain gauge.

**Important: Four Experiments should be added in above as per the requirement of the relevant subject.**

| RBM553 | MICROPROCESSOR LAB-I | L T P: 002 |
|---|---|---|

LIST OF EXPERIMENTS:

1.Study of 8085 Microprocessor kit.

2. Write a program using 8085 and verify for
a. Addition of two 8-bit numbers.
b. Addition of two 8-bit numbers (with carry).

3. Write a program using 8085 and verify for
a. 8-bit subtraction (display borrow)
b. 16-bit subtraction (display borrow)

4. Write a program using 8085 for multiplication of two 8- bit numbers by repeated addition or by bit rotation method.

5. Write a program using 8085 for division of two 8- bit numbers by repeated subtraction method and test for typical data.

6. Study of 8086 microprocessor kit.

7. Write a program using 8086 for finding the square root of a given number and verify.

8. Write a program using 8086 for copying 12 bytes of data from source to destination and

9. Write a program using 8086 and verify for:
a. Finding the largest number from an array.
b. Finding the smallest number from an array.

10. Write a program using 8086 for arranging an array of numbers in descending order.

11. Write a program using 8086 for arranging an array of numbers in ascending order and verify.

12. Write a program for finding square of a number using look-up table and verify. .

14. Write a program to control the operation of stepper motor using 8085/8086 Microprocessor and 8255 PPI.

| RBM554 | INTEGRATED CIRCUIT LAB | L T P: 002 |
|---|---|---|

1. Measurement of Op-amp Parameters. (Open Loop Gain, Input offset Voltage, CMRR, Slew rate)
2. Determination of Frequency response of Op-Amp.
3. Precision Rectifier
4. Instrumentation Amplifier.
5. Open Loop operation of Op-amp -Comparators - Schmitt Trigger.
6. Astable & Monostable Operation Using 555.
7. IC Voltage Regulator.
8. Voltage Controlled Oscillator.
9. Phase Locked Loop.
10. Frequency Multiplier
11. A/D Converters & D/A Converters.
12. Second Order Active Filter- High Pass & Low Pass Realization

**Important: Two Experiments should be added in above as per the requirement of the relevant subject.**

## DEPARTMENTAL ELECTIVE COURSE- 1

| REC501 | INTEGRATED CIRCUITS | L T P: 310 |
|---|---|---|

UNIT-I

**Analog Integrated circuit Design: an overview:** Current Mirrors using BJT and MOSFETs, Simple current Mirror, Base current compensated current Mirror, Wilson and Improved Wilson Current Mirrors, Widlar Current source and Cascode current Mirror.

**The 741 IC Op-Amp**: Bias circuit, short circuit protection circuitry, the input stage, the second stage, the output stage, and device parameters; DC Analysis of 741: Small Signal Analysis of input stage, the second stage, the output stage; Gain, Frequency Response of 741; a Simplified Model, Slew Rate, Relationship Between $f_t$ and SR

UNIT-II

**Linear Applications of IC op-amps:** An Overview of Op-Amp (ideal and non ideal) based Circuits V-I and I-V converters, generalized Impedance converter, simulation of inductors Filters: First and second order LP, HP, BP BS and All pass active filters, KHN, Tow-Thomas and State Variable Biquad filters; Sinusoidal oscillators

UNIT-III

**Digital Integrated Circuit Design-An Overview**: CMOS Logic Gate Circuits**:** 13.2-13.3 Basic Structure CMOS realization of Inverters, AND, OR, NAND and NOR Gates Latches and Flip flops**:** The Latch, The SR Flip-flop, CMOS Implementation of SR Flip-flops, A Simpler CMOS Implementation of the Clocked SR Flip-flop, D Flip-flop Circuits.

UNIT-IV

**Non-Linear applications of IC Op-amps:** Log–Anti Log Amplifiers, Precision Rectifiers, Peak Detectors, Simple and Hold Circuits, Analog Multipliers and their applications. Op-amp as a comparator, Zero crossing detector, Schmitt Trigger, Astable multivibrator, Monostable multivibrator, Generation of Triangular Waveforms.

UNIT-V

**D/A and A/D converters**

**Integrated Circuit Timer:** The 555 Circuit, Implementing a Monostable Multivibrator Using the 555 IC, Astable Multivibrator Using the 555 IC.

**Phase locked loops (PLL):** Ex-OR Gates and multipliers as phase detectors**,** Block Diagram of IC PLL, Working of PLL and Applications of PLL.

**Text Book:**

[1] Sedra and Smith, "Microelectronic Circuits", 4th Edition, Oxford University Press. Reference Books:
[2] Michael Jacob, `Applications and Design with Analog Integrated Circuits', PHI, 2nd Edn, 2006
[3] Jacob Milliman and Arvin Grabel, "Microelectronics", 2nd Edition, TMH, 2008.

| REC052 | Computer Architecture and Organization | L T P: 310 |
|---|---|---|

UNIT-I

Introduction to Design Methodology: System Design - System representation, Design Process, the gate level (revision), the register level components and PLD (revision), register level design The Processor Level:
Processor level components, Processor level design.

UNIT-II

Processor basics: CPU organization- Fundamentals, Additional features Data Representation - Basic formats, Fixed point numbers, Floating point numbers. Instruction sets - Formats, Types, Programming considerations.

UNIT-III

Data path Design: Fixed point arithmetic - Addition and subtraction, Multiplication and Division, Floating point arithmetic, pipelining.

UNIT-IV

Control Design: basic concepts - introduction, hardwired control, Micro programmed control - introduction, multiplier control unit, CPU control unit, Pipeline control- instruction pipelines, pipeline performance.

UNIT-V

Memory organization: Multi level memories, Address translation, Memory allocation, Caches - Main features, Address mapping, structure vs performance, System Organization: Communication methods- basic concepts, bus control. Introduction to VHDL.

**TextBooks:**

1. John P Hayes "Computer Architecture and Organisation", McGraw Hill Publication.

**Reference Books:**

1. M Morris Mano, "Computer System Architecture", Pearson Publication.
2. Carl Hamacher, Zvonko Vranesic and Safwat Zaky, "Computer Organization and Embedded Systems", McGraw Hill Publication.
3. David A. Patterson and John L. Hennessy, "Computer Organization and Design: The Hardware/Software Interface", Elsevier Publication.
4. I. Singh, Computer Organisation and Architecture, Khanna Publishing House

| RBM051 | HUMAN ANATOMY & PHYSIOLOGY | L T P: 310 |
|---|---|---|

**ANATOMY:**

**Unit 1**.

Structure & functions of cell. Polarization & depolarization of cell, Basic tissues & functions in brief.

**Unit 2**

**Outline of structures of the following systems:**

Cardiovascular system, Respiratory system, Alimentary system, Central Nervous system, Muscular System, Endocrine system Sense organs: Eye, Ear, Integument system (skin study)

**PHYSIOLOGY:**

**Unit 3**

Cardiovascular system,: Heart, conductive tissue of heart, cardiac cycle, heart valves, systemic &pulmonary circulation, Transmission of cardiac impulse, blood pressure. Respiratory system: respiration external (ventilation), Exchange in gases in the alveoli, Artificial respiration, Spirometer (Forced expiration volumes), peak flow meter. Alimentary system: all organs of the digestive system, other secretions & main functions.

**Unit 4**

Blood: composition of blood-blood cells & their functions. Cell counting, haemoglobin. Excretory system: Structure of Nephron, formation of urine & function of kidneys, urinary bladder, urethra, internal/external sphincters.

Nervous system: different parts, their functions, Reflex action & reflex arc. Function of sympathetic nervous system. Nervous conduction & action potentials.

**TEXT BOOKS:**

1. Anatomy and physiology in health and illness by : Ross and Wilson (ELBS pub)
2. Human Physiology by A. Vander, J. Sherman and D. Luciante.
3. Basic Human theory By Charles E Tobin Mc Graw Hill.

| REC054 | Artificial Neural Network | L T P: 310 |
|---|---|---|

UNIT-I

**Introduction to ANN:**

Features, structure and working of Biological Neural Network Trends in Computing Comparison of BNN and ANN.

Basics of Artificial Neural Networks - History of neural network research, characteristics of neural networks

terminology, models of neuron McCulloch - Pitts model, Perceptron, Ada line model, Basic learning laws, Topology of neural network architecture.

UNIT-II

**Back propagation networks :** (BPN) Architecture of feed forward network, single layer ANN, multilayer perceptron, back propagation learning, input - hidden and output layer computation, back propagation algorithm, applications, selection of tuning parameters in BPN, Numbers of hidden nodes, learning.

UNIT-III

**Activation & Synaptic Dynamics :**

Introduction, Activation Dynamics models, synaptic Dynamics models, stability and convergence, recall in neural networks. Basic functional units of ANN for pattern recognition tasks: Basic feed forward, Basic feedback and basic competitive learning neural network. Pattern association, pattern classification and pattern mapping tasks.

UNIT-IV

a)Feedforward neural networks -

- Linear responsibility X-OR problem and solution. - Analysis of pattern mapping networks summary of basic gradient search methods.

b) Feedback neural networks Pattern

Storage networks, stochastic networks and simulated annealing, Boltzmann machine and Boltzmann learning.

UNIT-V

**Competitive learning neural networks :**

Components of CL network pattern clustering and feature. Mapping network, ART networks, Features of ART models, character recognition using ART network.

Applications of ANN: Pattern classification - Recognition of Olympic games symbols, Recognition of printed Characters. Neocognitron - Recognition of handwritten characters. NET Talk: to convert English text

to speech. Recognition of consonant vowel (CV) segments, texture classification and segmentation.

**Text Book:**

1. B. Yegnanarayana, "Artificial neural Networks", PHI Publication.

**Reference Books:**

1. S. Raj Sekaran , Vijayalakshmi Pari," Neural networks, Fuzzy logic and Genetic Algorithms", PHI Publication.

2. Elaine Rich and Kevin Knight, "Artificial Intelligence", TMH Publication.

3. Rajiv Chopra, Machine Learning, Khanna Publishing House

| RBM601 | MICROCONTROLLER AND BIOMEDICAL APPLICATIONS | L T P: 300 |
|---|---|---|

**Unit 1**
**Introduction:** 8051, Comparison with microprocessor, pin diagram explanation, internal diagram 8051.

**Unit 2**
**Instruction Set:** Addressing mode, data transfer instruction, logical, arithmetic instruction, bit instruction, branching instruction.

**Unit 3**
**Timers:** Mode of timers, simple programming, generation of square wave.
**8051** connection to RS 232
**Interrupts:** Interrupt priority in 8051, generation of waveforms using interrupt, serial interface using interrupt.

**Unit 4**
**Interfacing** of memory, intelligent LCD, 8255, ADC, DAC, LED display.

**Unit 5**
**Applications:** Introduction to DSP processor, Applications of microcontrollers and computers in biomedical engineering, microcontrollers in embedded biomedical applications.

**Text Books:**
1. Micro controllers & its applications by B.S. Chhabra, Dhanpat Rai Pub. Co., India
2. 8051mC, Scott Mackenizie, PHI, Englewood Cliffs, New Jersey.
3. Myke Predko,'Programnming & Customizing the 8051 Microcontroller,' Tata McGraw-Hill Pub. Co. Ltd., New Delhi.

**Reference Books:**
1. 8051 m C Architecture Programming & Applications, K.J. Ayata: Penram International Publishers, India.
2. S.K. Venkata Ram,'Advanced Microprocessor & Microcontrollers, Luxmi Pub. Pvt. Ltd., New Delhi

| RBM602 | BIOMEDICAL SIGNAL PROCESSING | L T P: 310 |
|---|---|---|

**Unit 1.**
Z transform introduction, definition, convergence. Inverse Z transforms, Analysis of discrete time systems using Z transforms. Solutions of differential equations. Transfer functions and stability.

**Unit 2.**
Fourier transform for continuous signals. Energy spectrum, Properties (without proof), Gibbs phenomena, Auto and cross correlation. Discrete Fourier transforms. Properties (without proof), Inverse DFT, introduction to FFT

**Unit 3.**
IIR & FIR Filters, Low pass, High Pass, Band Pass Filters using windows – Kaiser Windows.
Sampling Theorem, aliasing Nyquist criteria, ADC's and DAC's.

**Unit 4.**
Digital signals and systems: Classification of systems causal, time varying, time invariant, lumped. Introduction to digital signals systems. Convolution, Auto-correlation and cross correlation , Use of Matlab signal processing toolbox on various real bio - medical signals**.**

**Unit 5.**
Introduction**,** Characteristics of Bio - Signals**,** Types of Signals**,** Measurement**,** Transformation. and reduction**,** computation of signal parameters that are diagnostically significant**,** stationary and non - stationary bio - signals**,** Application areas of Bio -Signals analysis - EEG, ECG, Phonocardiogram, Spiro Gram, Evoked Signals.

**TEXTBOOKS :**

1. Digital signal processing, Proakis (PHI)
2. Signal Analysis By R**.** P**.** Singh **,** Second edition Tata McGraw **–** Hill
3. Engineering Electronics By Mauro R Prentice – Hall
4. Malmivuo, J. and Plonsey, R. Bioelectromagnetism: Principles and Applications of Bioelectric and Biomagnetic Fields, Oxford University Press, New York, 1995.
5. D C Reddy, McGraw Hill, Biomedical Signal Processing.


**REFERENCES :**

1. Biomedical signal processing: Metin Akay (academic press)
2. Biomedical signal processing: Tompkins (academic press)
3. Theory and application of digital signal processing: Rabiner and Gold (EEE pub)

| RBM603 | BIOMATERIALS | L T P: 300 |
|---|---|---|

**Unit-1**

**Introduction:** Definition of biomaterials, requirements & classification of biomaterials, Comparison of properties of some common biomaterials. Effects of physiological fluid on the properties of biomaterials. Biological responses (extra and intra-vascular system). Surface properties of materials, physical properties of materials, mechanical properties.

**Unit-2**

**Metallic implant materials:** Stainless steel, Co-based alloys, Ti and Ti-based alloys. Importance of stress-corrosion cracking. Host tissue reaction with bio metal, corrosion behavior and the importance of passive films for tissue adhesion. Hard tissue replacement implant: Orthopedic implants, Dental implants. Soft tissue replacement implants: Percutaneous and skin implants, Vascular implants, Heart valve implants-Tailor made composite in medium.

**Unit-3**

**Polymeric implant materials:** Polyolefin's, polyamides, acrylic polymers, fluorocarbon polymers, silicon rubbers, acetyls. (Classification according to thermo sets, thermoplastics and elastomers).Viscoelastic behavior: creep-recovery, stress-relaxation, strain rate sensitivity. Importance of molecular structure, hydrophilic and hydrophobic surface properties, migration of additives (processing aids), aging and environmental stress cracking. Physiochemical characteristics of biopolymers. Biodegradable polymers for medical purposes, Biopolymers in controlled release systems. Synthetic polymeric membranes and their biological applications.

**Unit-4**

**Ceramic implant materials:** Definition of bio ceramics. Common types of bio ceramics: Aluminum oxides, Glass ceramics, Carbons. Bio resorbable and bioactive ceramics. Importance of wear resistance and low fracture toughness. Host tissue reactions: importance of interfacial tissue reaction (e.g. ceramic/bone tissue reaction).

**Composite implant materials**: Mechanics of improvement of properties by incorporating different elements. Composite theory of fiber reinforcement (short and long fibers, fibers pull out). Polymers filled with osteogenic fillers (e.g. hydroxyapatite). Host tissue reactions.

**Unit-5**

**Biocompatibility & Toxicological screening of biomaterials:** Definition of biocompatibility, blood compatibility and tissue compatibility. Toxicity tests: acute and chronic toxicity studies (in situimplantation, tissue culture, haemolysis, thrombogenic potential test, systemic toxicity, intracutaneous irritation test), sensitization, carcinogenicity, mutagenicity and special tests.

**TEXT BOOKS :**


1. Biomaterials Science**:** An Introduction to Materials in Medicine**,** By Buddy D**.** Ratner**,** et**.** al**.** Academic Press**,** San Diego**,** 1996**.**
2. Sujata V. Bhat, Biomaterials, Narosa Publishing House, 2002.
3. J B Park, Biomaterials - Science and Engineering, Plenum Press, 1984.

| RBM653 | Microprocessor Lab - II | L T P: 002 |
|---|---|---|

**LIST OF EXPERIMENTS**

1. Study of 8051 Microcontroller, Architecture & command.
2. Write an ALP for the Addition & Subtraction of 8 bit no's.
3. Write an ALP for multiplication of Two 8 bit no's.
4. Write an ALP for Division of Two 8 bit no's.
5. Write an ALP to find smallest & largest no in a given array.
6. Write an ALP to generate 10 KHz frequency using interrupt.
7. Write an ALP to interface intelligent LCD display with m C.
8. Write an ALP for m C & HLL for PC (VB/C++/VC++) to demonstrate/implement serial Interfacing.
9. Write an ALP to interface LED display.
10. Write an ALP to interface one m C with other using serial/parallel communication.
11. Write an ALP to switch ON alarm when m C receive interrupt

**Important: Three Experiments should be added in above as per the requirement of the relevant subject.**

| RBM652 | BIOMEDICAL SIGNAL PROCESSING LAB | L T P: 002 |
|---|---|---|

**LIST OF EXPERIMENTS**

1. Realization of signal-continuous & discrete by using MATLAB.
2. Write a MATLAB program to perform convolution of two signals.
3. Write a short program to perform
(a) DFT
(b) Inverse DFT
(c) FFT
By using MATLAB.

4. By using toolbox(MATLAB) simulate
(a) FIR Filter
(b) IIR Filter
5. Data acquisition of EEG & ECG signals by using DSP
kit. 6.Noise removal from EEG & ECG signals
7. Power spectrum analysis of EEG signals.

**Important: Five Experiments should be added in above as per the requirement of the relevant subject.**

| RBM651 | BIOINSTRUMENTATION LAB-II | L T P: 002 |
|---|---|---|

**LIST OF EXPERIMENTS**

1. To Study the Cardiovascular system.
2. Simulation of Cardiovascular system by using MATLAB/SIMULINK.
3. To Study the Heart Model and simulate it using MATLAB/SIMULINK.
4. To Study the Eye Movement System, its mathematical mode.
5. To study linear muscle model.
6. To study model of respiratory mechanics.
7. Implement the simulink model for Lung Mechanics.
8. Implement the glucose insulin regulation model by MATLAB tools.
9. To study the circulatory model by MATLAB.
10. Implement the simulink model for neuromuscular transient response

**Important: Four Experiments should be added in above as per the requirement of the relevant subject.**

## DEPARTMENTAL ELECTIVE COURSE- 2

| RBM061 | LASER & FIBER OPTICS AND ITS MEDICAL APPLICATION | L T P: 310 |
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**Unit1.**
Introduction to fiber optics: Basic fiber link, applications, principles of light: Introduction, EM spectrum, internal & external reflections, Snell' slaw, optical fiber numerical aperture, Fresnel reflection.

**Unit2.**
Optic fiber & its properties: Introduction, Basic fiber construction, propagation of light, modes of operation, refractive index profile, types of fibers, dispersion, data rate and bandwidth, attenuation, losses. Connectors, Splices & Couplers: Introduction, splices: mechanical, fusion, protection of splice, connectors: SMA, STC, bionic etc, coupling: passive, Stan, TEE types. Optical sources & Photo Detectors: Introduction: creation of photons, LED, ILD, photo detectors: introduction, PIN photodiode, avalanche photodiode, photodiode parameters, detector noise, speed of response, SNR.

**Unit 3.** Modulation scheme for fiber optics transmission: Introduction, digital modulation, analog modulation schemes, multiplexing.

**Unit 4.**
Laser Systems: Introduction, types of lasers: Solid state lasers, Gas lasers, Dye lasers, Lasers used in medical practice: Ruby laser, CO2 laser, Nd-Y AG laser and related solid state laser. Laser -Tissue Interaction: Terminology : spectral band designations, energy & power, irradiant & radiant exposure, fluency, thermal diffi1sion fibers & contact tips, Types of laser-tissue interactions

**Unit 5.** Laser Application in Medical Therapy: Introduction, application in general surgery, dermatology, ophthalmology, cardiovascular & chest surgery, dentistry, neuro surgery, otolaryngology & head and neck surgery, tumor surgery, gynecologic laser.

**TEXT BOOKS :**


1. Therapeutic Lasers -Theory and practice by G. David Baxter, Churchill Living stone publications.
2. Medical Lasers and their safe use by David H Shiney, Stephen and L. Trokel, Springer-Verlag publications.
3. Elements of fiber optics by S. L. Wymer, Regents-Prentice Hall publications.
4. Biomedical Electronics & Instrumentation by S. K. Venkata Ram, Galgotia publications.


**REFERENCES:**


1. Laser and optical fibers in medicine by Katzer and Abraham, Academic press publications
2. An Introduction to optical fibers by A. M. Cherin, McGraw Hill publications.

| RBM062 | BIOELECTRICITY | L T P: 310 |
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UNIT-I

Bioelectricity generation at the cellular & sub cellular level. Different biopotentials and their characteristics.

UNIT-II

Nernst Equation: Derivations and its significance. Refractory Period Characteristics of Stimulus. Strength-Duration relationship. Electrical equivalent circuit of Axon. Membrane time and space constants.

UNIT-III

Hodgkin-Huxley formulation, Membrane conductance, Nerve conduction, membrane properties from current voltage relations, Models of squid axon. Propagation of impulses in unmyelinated and myelinated nerve fiber. Electrical properties of receptors. Intensity-frequency relationship. Electrical properties of synaptic junctions - EPSP and IPSP.

UNIT-IV

Characteristics of Action potentials at SA Node, Atria, A V Node, Purkinje fibers and Ventricles. ECG Complexes. 12 lead ECG. Standard leads of Einthoven. Pericardial leads and Augmented limb leads. Relationship between unipolar extremity leads and standard Bipolar leads. Electrical activity of skeletal muscles, Motor unit potentials, neuromuscular transmission, EMG wave form.

UNIT-V

Biopotential electrodes: classification & characteristics. Electrode-Electrolyte Interface, Equivalent Circuit Properties of Needle & Micro Electrodes, Electrodes for Surgery, Physiotherapy & Analytical instruments.

**TEXT BOOKS**

1. Robert Plonsey and Roger Barr, Bioelectricity, McGraw Hill, 1986.
2. John Webster. Medical Instrumentation.- Application and Design. John Wiley and Sons. Inc., New York. Third edition 2003.

**REFERENCES:**

1. L.A Geddes, Principles of Applied Biomedical Instrumentation, John Willy & Sons, 1989.
2. Plonsey Robert and Flemming David G. Bioelectrical phenomena, McGraw Hill, 1969.

| RBM063 | BIOMECHANICS | L T P: 310 |
|---|---|---|

**Unit 1. Scalar And Vector Quantities:**
Different operations on vectors, forces and Moments.

**Unit 2.**
System of Forces in 2D and 3D; Equilibrium equation, Applications with examples on Human Body.

**Unit 3**
Work Energy Equation, Application to Bio-Medical System.

**Unit 4.**
Stress Strain Diagram, Stress in Bending ,Torsion and Compound Loading, Stress Shielding of Bone.

**Unit 5.**
Mechanical Properties of Human Bone and Soft Tissues, Cortical and Cancellous Bone, Viscoelasticity, Elastic Model of Bone.

**TEXT BOOKS :**
1.Ozkaya Nihat,Margrate Nordine,"Fundamental of Bio- Mechanics". Springer Publication.
2.Lucas G. L.,W.F.Cook " A priemer of Bio Mechanics". Springer Publication.
3.Gardiner M .Dena ," The Principle of Exercise Therapy". CBS Publicer.c. Orthotics

| RBM064 | PHYSIOLOGICAL CONTROL SYSTEM AND SIMULATION MODELLING | L T P: 310 |
|---|---|---|

**Unit 1**.

Introduction to state variable analysis of control systems: -Introduction to state variable concept, definition of state variables, matrix representation of state equation, state transition equation, properties of transition matrix, relationship between state equations and higher order differential equations, state equation and transfer function, characteristics equation, Eigen values & Eigen vectors.

**Unit 2**.

Transformation to phase variables canonical forms of state variables, controllability canonical form, observability canonical form Jordan canonical form, controllability of linear system, observability of linear system relationship among controllability, observability and transfer function.

**Unit 3**.

Introduction to biological control system: Introduction, Dynamic systems and their control, modeling and block diagrams, the pupil control systems(Human Eye), general structure of control systems, the dynamic response characteristics of the pupil control system, open & close loop systems instability, automatic aperture control.

**Unit 4**.

Mathematical modeling of the system: Thermo regulation, Thermoregulation of cold bloodedness & warm bloodedness, the anatomy of thermo regulation, lumping & partial differential equations, heat transfer examples, mathematical model of the controlled process of the body.

**Unit 5**.

Modeling the body as compartments, behavior in simple compartmental system, pharmacokinetic model, multi compartmental system. distribution and accessibility of body water & tissue compartments, basis for zero order & first order chemical kinetic behavior in the biological system.

**Unit 6**.

Biological receptors: -Introduction, receptor characteristics, transfer function models of receptors, receptor and perceived intensity. Respiratory model & systems, Neuromuscular model, Cardiovascular control system.

**TEXTBOOKS :**


1. Automatic control systems: By Benjamin C Kuo.
2. Control system Engineering: By I. J . Nagarath. & M. Gopal.
3. Bio- Medical Engineering Principles By: David. 0. Cooney , Michel Deckker INC
4. Biological control systems: John H Milsum Mc Graw Hill 1966.
5. The Application Of Control Theory Of A Physiological System by Howard T ilhorn Sounders Publication